

SOFTWARE REQUIREMENTS SPECIFICATION

Counselling Services Appointment Manager *and Digital Communication App*

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STAKEHOLDER

Wellness Centre,
Central University of
Technology, Free State



SUBJECT

ITC327W —
Work-Integrated
Learning



ASSESSMENT

Group Assessment 1
*Phase 1: Planning,
Requirements, and Feasibility*



DATE

August 2026



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1. Project Introduction

This document is the Software Requirements Specification (SRS) for the Counselling Services Appointment Manager and Digital Communication App, developed by Group D as part of the ITC327W Work-Integrated Learning module. It combines and refines the work completed in Units 1 and 2, tracing the project from the initial identification of the stakeholder's problem through to a realistic, agreed project baseline that will carry forward into system design.

The stakeholder for this project is the Wellness Centre, which provides psychological counselling services to students. The Centre currently manages appointment bookings, student records and statistical reporting using a combination of paper records, the Outlook calendar, and Excel spreadsheets. This SRS documents the problem investigated with the stakeholder, the requirements derived from that engagement, and the scope of the proposed solution — a Flutter mobile application and an ASP.NET Core web application, sharing a single Supabase backend.



2

Stakeholder Profile and Current Process



2.1 Organisation and Stakeholder

Organisation: The Wellness Centre, Central University of Technology, Free State.

- **Admin / Front Office:** the first point of contact for students; records appointment requests and manages the triage log.
- **Manager and Psychologist:** conducts therapeutic sessions and oversees service delivery, including stepping in as a psychologist when required.
- **Intern Psychologist:** assists with counselling sessions under supervision.



2.2 Affected Users

- Students who request and attend counselling appointments.
- Wellness Centre administrative staff who process bookings and maintain records.
- Psychologists and counsellors who deliver sessions and report on statistics.



2.3 Current Process

The Wellness Centre relies on a mixture of manual and electronic processes:

- **Appointment booking:** students request appointments either in person at reception or, most commonly, via the Outlook calendar.
- **Emergency walk-ins:** attended without a booking, but only when the situation is judged sufficiently serious.
- **Student information:** captured and maintained using Excel spreadsheets, with statistics generated per psychologist.
- **Triage:** first-time students complete a paper triage log before seeing a counsellor, no separate register is kept, in order to protect confidentiality.



2.4 Identified Problems

- ✗ Paper records are difficult to manage and not secure.
- ✗ Outlook calendar causes scheduling conflicts and is not integrated with records.
- ✗ Excel spreadsheets are manual, time-consuming and prone to errors.
- ✗ Lack of real-time statistics and reporting.
- ✗ No central system for student information, appointments and records.
- ✗ Confidentiality risks with paper-based triage.



3.

Problem Statement



3.1 Current Situation

The Wellness Centre currently depends on the Outlook calendar and paper-based methods to record appointment requests. The admin reviews and approves each request and assigns a psychologist, while student records and statistics are maintained separately in Excel by each psychologist.



3.2 The Problem

This manual, fragmented process leads to missed appointment requests, double bookings, misplaced student records, and delays in generating accurate statistics. Emergency cases are only prioritised when judged sufficiently serious, and there is no reliable way to search existing student records.



3.3 Affected Users

- Students — experience missed or unavailable appointments and delays.
- Psychologists — struggle to balance client sessions with manual statistics capture.
- Admin staff — carry the full burden of manually tracking bookings and records.



3.4 Consequences

- Reduced trust in the booking process and inconsistent access to counselling support.
- Increased administrative workload and risk of human error.
- Incomplete or delayed statistics and reporting to Wellness Centre management.



3.5 Need for Improvement

The Wellness Centre requires a centralised system that allows students to book appointments online, securely manages student records through appropriate access control and encryption, is user-friendly, generates statistics automatically, and gives the Centre a platform to communicate programmes and events to students.



4.

Project Aim and Objectives



4.1 Purpose

The purpose of the Counselling Services App is to provide a secure, reliable and user-friendly platform for managing student counselling services, reducing the problems experienced with the current paper-based and Outlook-based process.



4.2 Aim

The aim of the project is to improve the management and delivery of student counselling services by replacing the current paper-based and Outlook-based processes with a centralised digital system that is secure, efficient, accessible and user-friendly.



4.3 Objectives

- Allow students to request and manage counselling appointments online.
- Reduce missed appointment requests and double bookings.
- Allow Wellness Centre staff to efficiently manage and track appointments.
- Securely store and manage student information and records.
- Make student records easier to search, access and manage.
- Allow psychologists to manage their appointments and client information.
- Generate accurate statistics and reports without requiring manual compilation.
- Improve communication between students and Wellness Centre staff.
- Provide information about Wellness Centre programmes, activities and events.
- Protect sensitive student information through appropriate security and encryption.

5.

Project Scope



5.1 Deliverables

- Requirements Specification — functional and non-functional requirements of the system.
- System Design — UML diagrams, database design, user interfaces and system architecture.
- User Interface Prototype — the main screens and functionality of the application.
- Functional Counselling Services App — a working application providing the required functions.
- Appointment Management — requesting, approving, scheduling and tracking appointments.
- Student Information Management — secure storage, updating, searching and management of records.
- Statistics and Reporting — generation of relevant statistics and reports.
- Communication and Announcements — publishing Wellness Centre programmes and events.



5.2 Acceptance Criteria

- Students can register, log in and securely access the application.
- Students can submit counselling appointment requests.
- Wellness Centre staff can approve, reject, reschedule and manage appointment requests.
- The system prevents or identifies double bookings.
- Psychologists can view their assigned appointments and relevant student information.
- Student records can be securely stored, updated, searched and retrieved.
- Sensitive student information is protected through appropriate security and encryption.
- The system can generate accurate statistics and reports.
- Wellness Centre staff can create and publish announcements about programmes and events.
- Different users have access only to functions and information appropriate to their role.
- The application is easy to navigate and understandable for its intended users.
- Testing confirms that the main system functions work according to specification.



5.3 Exclusions (Out of Scope)

- Staff counselling appointments (staff use the institution's K-Ways system).
- External (non-student) counselling.
- Replacing the institution's K-Ways system.
- General medical services unrelated to counselling.
- Providing automated psychological treatment.
- Replacing the professional judgement of psychologists or counsellors.
- Generating statistical graphs.

5.

Project Scope



5.4 Constraints

- The system must protect confidential counselling information.
- Access must be restricted to authorised users.
- The system must primarily support student counselling services.
- Emergency walk-ins must be handled differently from normal appointments.
- The system must accommodate the counselling service's existing triage process.
- The system must prevent double bookings.
- The system must support the reporting requirements of the counselling service.



5.5 Assumptions

- Students will have access to the application using a suitable device.
- Students will have valid student numbers.
- Counselling staff will have individual system accounts.
- The institution will provide the necessary infrastructure for hosting and maintaining the system.
- The institution will determine which staff members are authorised to access sensitive information.
- Internet access will generally be available for virtual counselling sessions.
- The system will need to comply with the institution's privacy and information-security requirements.
- Exact counselling categories and statistical fields will be confirmed by the counselling service.

6. Stakeholder Needs

Stakeholder needs were gathered through a structured interview with the Wellness Centre admin, manager and psychologist, supported by a short student survey on booking preferences. The full interview questions, responses and survey results are provided as evidence in Appendix A.



6.1 Key Findings

- The Outlook calendar is the most frequently used booking channel today, but requests are sometimes missed.
- Only one full-time psychologist is available, supported by the Manager and an intern psychologist when needed.
- Required booking information: student number, name and surname, and contact details.
- No sign-in register is kept for confidentiality reasons; students instead complete a triage log.
- Statistics reported to management include attendance numbers, common presenting issues, gender, and vital signs (blood pressure).
- Only admin and psychologists need access to booking details.
- The stakeholder wants a secure, encrypted, user-friendly system that is not hosted on shared consumer storage (e.g. OneDrive).
- Walk-ins are accepted only for genuine emergencies; students are otherwise expected to book in advance.



6.2 Confirmation

The stakeholders (Wellness Centre admin, manager and psychologist) confirmed that the above findings accurately reflect their needs and expectations. They agreed that the proposed system should address these needs and support more efficient, secure and reliable management of counselling services.

ID	Requirement	Description
FR-01	User Authentication	The system must allow authorised users to log in securely.
FR-02	Student Appointment Booking	The system must allow students to request counselling appointments online.
FR-03	Appointment Management	The system must allow admin staff to view, create, update, approve, reschedule and cancel appointments.
FR-04	Double-Booking Prevention	The system must prevent or flag double bookings of an appointment slot.
FR-05	Psychologist Appointment View	The system must allow psychologists to view their assigned appointments.
FR-06	Student Record Storage	The system must store student number, name, contact details and assigned psychologist.
FR-07	Record Search	The system must provide a search facility for authorised staff to locate student records.
FR-08	Digital Communication	The system must allow students to communicate virtually with a counsellor where appropriate.
FR-09	Appointment Status Updates	The system must allow the psychologist to update the status of an appointment.
FR-10	Role-Based Access Control	The system must restrict access to information and functions based on the user's role.
FR-11	Statistics Generation	The system must generate statistics and reports without requiring manual compilation.
FR-12	Announcements	The system must allow the Wellness Centre to publish programme and event announcements to students.



8.

Non-Functional Requirements

ID	Category	Requirement
NFR-01	Security	The system must protect confidential student and counselling information from unauthorised access, using encryption where appropriate.
NFR-02	Usability	The system must have a clear, user-friendly interface suitable for students, admin and psychologists.
NFR-03	Performance	The system should retrieve student information and appointment data quickly.
NFR-04	Availability	The system should be available whenever students and staff need to access it.
NFR-05	Reliability	The system must reliably maintain student records and appointment bookings without loss of data.



9.

Integration Requirements

The proposed system consists of a web application, a Flutter mobile application, an ASP.NET Core backend, and a shared Supabase (PostgreSQL) database. The integration requirements below describe how these components will operate as a single, consistent solution.

Integration Requirement
The web application and Flutter mobile application shall connect to the same backend services and shared database, so information created or updated on one platform is available on the other.
The web and mobile applications shall use the same authentication mechanism so users can securely access the system from either platform.
The applications shall communicate with the backend using secure HTTPS connections.
The system shall integrate with Supabase (PostgreSQL) to store and retrieve student, appointment, psychologist and counselling-session data.
Changes made to an appointment through the web application shall be reflected when the same appointment is viewed through the mobile application, and vice versa.
The system shall provide consistent appointment status information across the web and mobile applications.
If virtual counselling is implemented, the system shall integrate with an approved video-conferencing service to support virtual sessions.
The web and mobile applications shall apply the same validation and business rules to prevent problems such as double bookings and invalid appointment information.

10.

Requirements Change Log

Use this log to record every change to the agreed requirements/scope after the initial SRS was signed off — additions, modifications, removals, or clarifications. Log a change as soon as it's requested, before it's implemented, so the team and the Wellness Centre stakeholder always have a shared record of what changed and why.

Sample entries below illustrate how the log is used, based on plausible change requests arising from the Initial SRS. Replace or extend these with the team's actual change history as the project progresses.

Change ID	Date Raised	Requirement Affected	Change Type	Description of Change	Reason / Justification	Requested By	Status
CR-001	2026-03-10	Communication and Announcements	Addition	Added requirement for students to view Wellness Centre programme and event announcements within the app, not just receive them via email.	Wellness Centre admin requested an in-app channel to raise awareness of upcoming events, since email announcements were often missed by students.	Wellness Centre Admin	✔ Implemented
CR-002	2026-03-24	System Requirements — Virtual Counselling	Clarification	Confirmed virtual counselling integration is conditional on the Wellness Centre approving a specific video-conferencing platform (moved from "Must Have" to "Could Have" in scope).	SRS wording only required virtual counselling support. "If implemented" — no platform had been approved yet, so it could not be treated as essential.	IM Model	✔ Approved
CR-003	2026-04-02	Statistics and Reporting	Removal	Removed generation of statistical graphs from the deliverable; system will produce raw statistics/reports only, not visual charts.	Wellness Centre stakeholder confirmed they only need the underlying numbers for their own reporting process and do not require built-in graphing.	Wellness Centre Admin	✔ Implemented
CR-004	2026-04-15	Appointment Management — Emergency Requests	Addition	Added a distinct emergency walk-in request pathway separate from the normal scheduled booking flow, to match the counselling service's existing triage process.	Counselling staff explained that emergency cases must be flagged and prioritized differently from standard bookings, which the original booking flow did not support.	TC Selape	✔ Implemented
CR-005	2026-04-28	System Requirements — Double Booking Prevention	Modification	Extended double-booking prevention to check across all psychologists sharing a timeslot, not just the psychologist being booked.	Testing revealed that two students could still be booked into overlapping slots if assigned to different psychologists who share a room/resource.	N Dlal	✔ Implemented
CR-006	2026-05-06	Requirements Prioritization — Notifications	Modification	Proposed moving appointment reminder notifications from "Should Have" to "Must Have" ahead of the next milestone.	Early user feedback from psychologists indicated missed appointments were high occurrence without reminders, undermining a core project objective.	T Mahlangu	🕒 Under Review



10.

Requirements Change Log

Change ID	Date Raised	Requirement Affected	Change Type	Description of Change	Reason / Justification	Requested By	Status
CR-007	2026-05-14	Student Information Management — Search	Modification	Requested extending the student record search facility to allow filtering by assigned psychologist, in addition to student number and name.	Admin staff noted that searching by psychologist would speed up caseload reviews during statistics compilation.	BBL Ntsutle	Under Review
CR-008	2026-05-20	Out of Scope — Staff Counselling	Clarification	Re-confirmed in writing that staff counselling appointments remain fully out of scope and stay on the institution's K-Ways system.	A staff member queried whether the new app also covered staff bookings; needed a documented answer to avoid scope confusion going forward.	Wellness Centre Admin	Approved
CR-009	2026-05-29	Role-Based Access Control	Addition	Requested a new "Assistant Admin" access role with limited permissions (view/search only, no record editing) for reception staff.	Reception staff need to look up appointment status for students at the front desk but should not be able to edit confidential records. Rejected: adding a fourth role was judged out of scope for the current timeline and can be revisited post-handover.	F Mahlumba	Rejected
CR-010	2026-06-05	Requirements Prioritization — Psychologist Availability	Addition	Added requirement for psychologists to set and update their own available time slots, rather than admin manually managing every psychologist's calendar.	Admin raised that manually tracking six psychologists' availability was becoming a bottleneck and a likely source of scheduling errors.	IM Molefi	Approved
CR-011	2026-06-12	Communication and Announcements — Wellness Tips	Addition	Added an optional section for the Wellness Centre to publish short wellness tips and educational content alongside event announcements.	Wellness Centre staff felt the awareness objective in the SRS was better served by ongoing tips, not just one-off event posts.	T Mofokeng	Logged
CR-012	2026-06-19	Security — Encryption of Student Information	Clarification	Clarified that encryption applies to student records and counselling-session data at rest in Supabase, and to all data in transit over HTTPS, in line with the institution's privacy requirements.	The SRS stated encryption was required but did not specify at-rest vs in-transit; the institution's IT policy needed this made explicit before the data model was finalized.	Wellness Centre Admin	Approved



10. Field Guide

Change Type

one of: Addition, Modification, Removal, Clarification.

Status

one of: Logged, Under Review, Approved, Rejected, Implemented.

Requirement Affected

reference the specific SRS section or requirement (e.g. "Statistics and Reporting", "System Requirements — double booking").

Every approved change

should be reflected back into the SRS and the Refined Project Scope document so both stay current.

11.

Refined Project Scope

1. Scope Statement

The Counselling Services Appointment Manager and Digital Communication App is a centralized web and Flutter mobile system that will replace the Wellness Centre's current paper-based and Outlook-based appointment process and Excel-based statistics. The system will let students request and manage counselling appointments, let administrators and psychologists manage appointments and student records securely, and let the Wellness Centre communicate programme and event information to students. The web and mobile applications share one ASP.NET Core backend and one Supabase (PostgreSQL) database, so data stays consistent across both platforms.

2. In-Scope Functional Areas

- Appointment Management — requesting, approving, scheduling, rescheduling, cancelling, and tracking counselling appointments, with double-booking prevention.
- Student Information Management — secure storage, updating, searching, and retrieval of student records.
- Psychologist/Counsellor Functions — viewing assigned appointments, relevant student information, and updating appointment status.
- Statistics and Reporting — automatic generation of counselling statistics without manual compilation by psychologists.
- Communication and Announcements — Wellness Centre-to-student messaging about programmes, activities, and events.
- Security — authentication, role-based access control, and encryption of confidential student information.
- Cross-Platform Consistency — shared backend and database so actions on web or mobile are reflected on both.

3. Requirements Prioritization (MoSCoW)

Refined from the group's SRS to remove duplication and align "Could Have" virtual counselling with its actual system constraint (implementation is conditional on stakeholder approval of a video platform).



Must Have — essential for the system to function

- Student registration and secure login
- Appointment booking and scheduling (with double-booking prevention)
- Appointment approval and management by admin (confirm, cancel, reschedule)
- Assignment of students to psychologists/counsellors
- Student information and record management (store, update, search, retrieve)
- Secure storage and encryption of student information
- Role-based access control (Students, Administrators, Psychologists/Counsellors)
- Appointment tracking and status updates
- Statistics generation (non-graphical)
- Basic communication between students and Wellness Centre staff



Should Have — important, included if time and resources allow

- Appointment reminders and notifications
- Emergency / urgent case (walk-in) requests
- Reports
- Psychologist availability management
- Viewing appointment history
- Notifications about appointment changes

11.

Refined Project Scope

3. Requirements Prioritization (MoSCoW)



Could Have — improves the system but not essential

- Student feedback and ratings
- Wellness Centre programme and event announcements
- Wellness tips and educational information
- Additional communication features
- Customizable user notifications
- Virtual (video) counselling — conditional on stakeholder approval and an approved video-conferencing platform



Won't Have — excluded from this version

- Medical diagnosis
- Prescribing medication
- Processing medical aid claims
- Managing external hospital records
- Replacing professional psychologists, counsellors, or face-to-face sessions
- Staff counselling appointments (handled via the institution's K-Ways system)
- External (non-student) counselling
- Generating statistical graphs

11.

Refined Project Scope

4. Out of Scope

- Staff counselling appointments (remains on the institution's K-Ways system).
- External (non-student) counselling.
- Replacing the institution's K-Ways system.
- General medical services unrelated to counselling.
- Automated psychological treatment or diagnosis.
- Replacing the professional judgement of psychologists/counsellors.
- Generating statistical graphs (raw statistics generation is in scope; graphing is not).

5. Key Assumptions Affecting Scope

- Students have access to a suitable device and a valid student number.
- Counselling staff will have individual system accounts.
- The institution provides hosting/infrastructure and determines which staff may access sensitive information.
- Internet access is generally available, including for any virtual counselling sessions.
- Exact counselling categories and statistical fields will be confirmed by the counselling service.

6. Scope Boundaries / Constraints

- The system must protect confidential counselling information and restrict access to authorized users only.
- The system must primarily support student counselling services, not staff counselling.
- Emergency walk-ins must be handled differently from normal scheduled appointments.
- The system must accommodate the counselling service's existing triage process.
- The system must prevent double bookings and support the counselling service's reporting needs.

Appendix A — Stakeholder Evidence

This appendix contains supporting evidence gathered from stakeholders during Units 1 and 2, confirming the problems, needs, and requirements documented in this SRS.



A.1 Meeting Request and Stakeholder Response

Proposal & Appointment Request: Software Development Collaboration for Campus Wellness

 Summarize this email



Dlali Ntlahla



Reply



Reply all



Forward



To:  Mpuru Mathabo

Mon 17 Aug 2026 11:40

Good Day

I hope this email finds you well.

We are writing on behalf of a student software development team (DevOps) here at the Central University of Technology (CUT).

We would like to meet with the Counselling services administrator to share some ideas and implement an interview based on the current process of the services, problems experienced and an opinion of the admin in terms of the services.

We would like to schedule the meeting for Thursdays and Wednesdays, since that's the time the team is available on campus.

We look forward to hearing from you.

Kind regards,

Proposal & Appointment Request: Software Development Collaboration for Campus Wellness Summarize this email



Mpuru Mathabo



 Reply

 Reply all

 Forward



To:  Dlali Ntlahla

Mon 17 Aug 2026 12:26



You replied on Mon 17 Aug 2026 12:29

Dear Ntlahla

Hope you are well

Please specify if the meeting would be on both days Wednesday and Thursday or you just need one day and specify your time of availability.

Thank you

Regards



Central University of
Technology, Free State

Mathabo Mpuru

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Welcome back to the new Outlook (1/5)



INTERVIEW QUESTIONS & STAKEHOLDER RESPONSES.

1. Regarding the counselling services, what role do you play and how do you interact with clients?

Admin: Front office, students have the first interaction with the admin, and they sign a triage Log on the date of the appointment. Manager&Psychologist: The interaction I have with clients, is mostly therapeutic sessions

2. Currently, how do the clients book for appointments?

We have a booking system(on paper) we use when a client comes manually, or a client can use outlook to request an appointment date. The one we use the most is the Outlook calendar.

[Follow Up Question]

- 2.1. Are the clients only the students, or external students and staff can request appointments too?

We don't accept requests from external students, and for the staff, they have their own called K-ways*.

- 3.How do you record and manage the student information?

Currently, we use record the student information and statistics from the appointment requests, and we generate stats per psychologist. We use excel spreadsheet to record statistic.

[Follow Up Question]

3.1. So how many psychologist do you have for the services?

We only have one psychologist, however the Manager can step in as a psychologist too, also with the intern psychologist.

4.What problems do you often face with the current process you're working with?

Emails are missed, because sometimes we may see emails from bottom to top, unless you send a feedback email. Sometimes a student may come for a session and there will be no one to attend them. In terms of statistics we see clients each and every hour, so we might slag at taking stats.

5.How often does the issue of double bookings, missed appointments, or lost records happen?

This happens often. Students records can even be misplaced sometimes, and because of the disadvantage of not having a searching system. Even sometimes it can happen that a student won't remember their psychologist

[Follow Up Question]

5.1. How does it resolved?

If maybe a student will attend a session and it happens we lost or misplaced the file, we would assign them to an available psychologist.

6. What specific information do you require when they book for an appointment?

Student Number #Name and surname # Contact details We also require them because, if there's an email reminder, we can contact a student.

[Follow Up Question]

6.1. Is there any register that a student needs to sign to see the doctor?

No, for confidentiality, we keep no register, but what we do is to make student sign a triage Log before seeing the Counsellor?

7. What data or summaries do we need to report the manager?

it's the: -How many students attended sessions -Which diagnosis is much prominent at the time of the session(HIV AND AIDS, PSYCHOLOGIST, PERSONAL PROBLEMS, ETC...) -Gender - If their BP or LH is high or low

8. Who on the team should have access to view and edit booking details? -Obviously, the admin.

9. Besides, who else will be interacting with the system?

The psychologist, and the admin too.

10. If you had the power to change the system, how would you want to work?

should be user-friendly, meaning it should be able to dissect data, pull out data, generate statistical graphs, it should be secure and encrypted. We don't portal systems on One Drive because it can be accessed by anyone.

[Requirements Checker Follow Up Question]

10.1. Do walk-ins happen normally or do we have to book for an appointment first?

For walk-ins we usually attend emergencies only, and if your situation is serious. However, a student should book an appointment first.

10.2. Are most students comfortable with face-to-face sessions?

Not all students, but what we understand as psychologists, we need to build rapport first plus some students may be reluctant at the beginning of the sessions.

10.3. Is there any difference in human connection virtually and face to face?

Yes, because they may be problems with the network where student is at, and as much as we can use hybrid sessions IT IS STILL DANGEROUS.

11.(From the Manager to us) "Have you guys been to therapy before?"

us: yes, that's why we're willing to help others too.

A.3 Meeting Notes

1. Front office in terms of booking

Back - awareness to come up with a prevention approach

2. Use an outlook calendar for only students

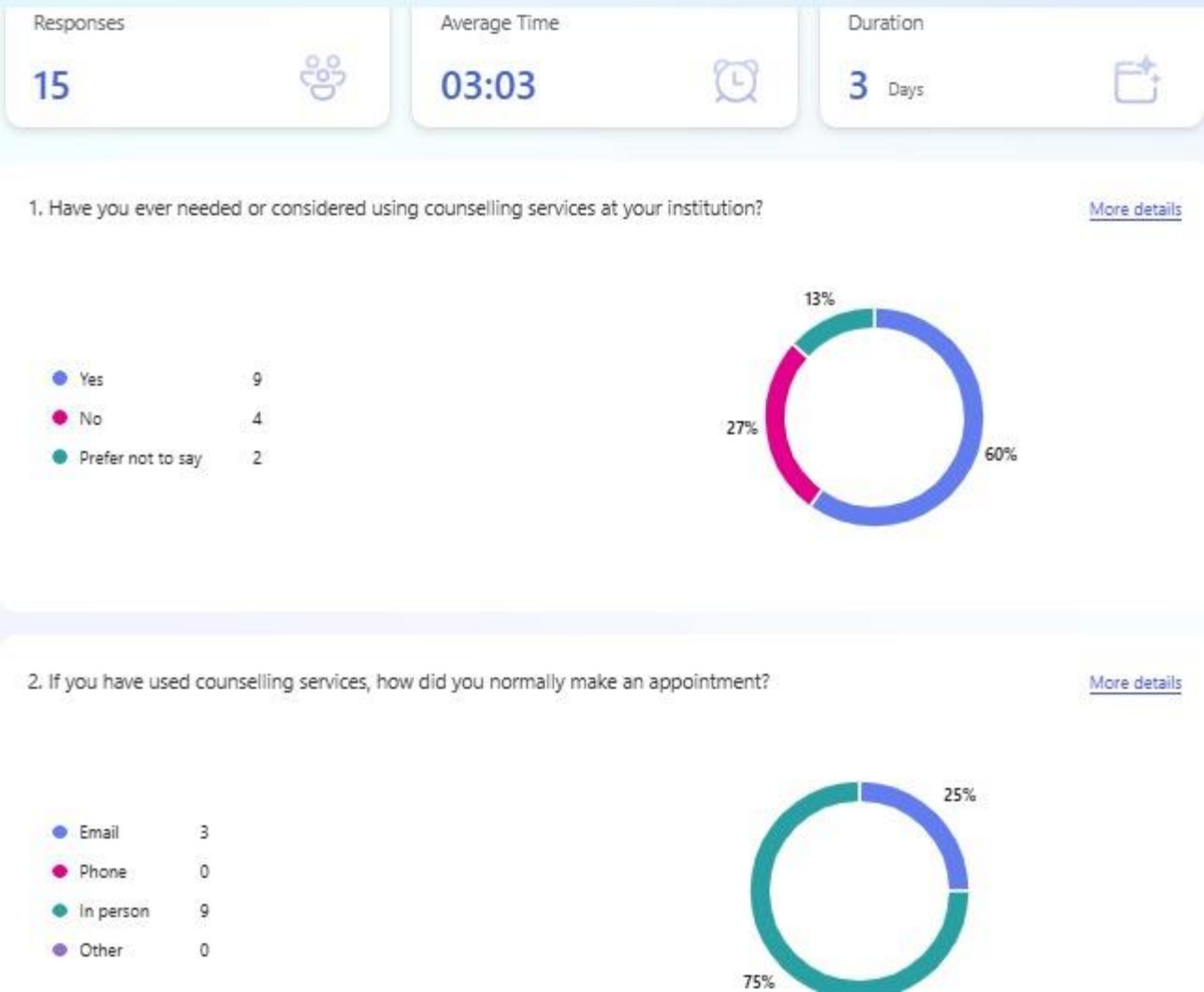
Generate stats indivi among 3 proffs
on an excel spreadsheet

- Emails are missed unless feedback
- Not always someone avail
- Might lag due to overwhelm
- Spreadsheet not secure
- Double booking are an issue
- Records are often missed (No numerical system)

Issue is stats

Admin must be able to edit

A.4 Students Feedback



1. Have you ever needed or considered using counselling services at your institution?

[More details](#)

Yes

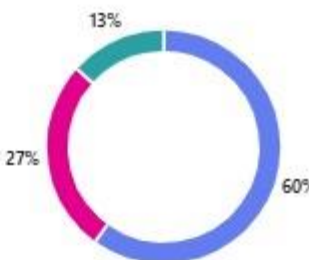
9

No

4

Prefer not to say

2



Response	Count	Percentage
Yes	9	60%
No	4	27%
Prefer not to say	2	13%

2. If you have used counselling services, how did you normally make an appointment?

[More details](#)

Email

3

Phone

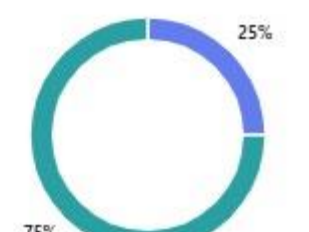
0

In person

9

Other

0

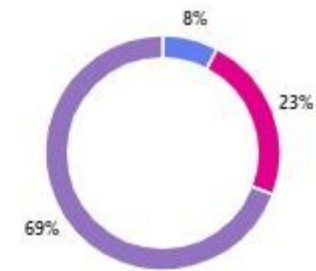


Response	Count	Percentage
Email	3	25%
Phone	0	0%
In person	9	75%
Other	0	0%

3. What difficulties, if any, have you experienced when trying to access counselling services?

[More details](#)

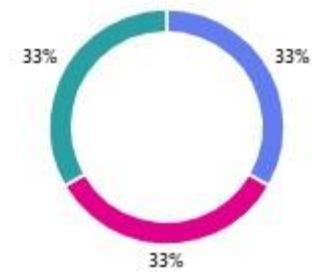
● Difficulty of getting an appointment	1
● Long waiting times	3
● Lack of privacy	0
● Not knowing if the counsellor is available	9



4. What is the biggest barrier that prevents you from using counselling services?

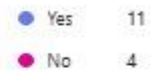
[More details](#)

● Fear of being Judged	5
● Privacy concerns	5
● Not knowing about them	5
● Other	0



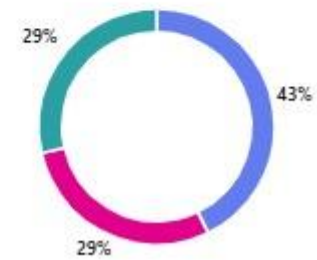
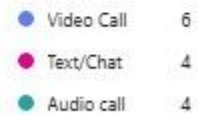
5. Would you be interested in communicating with a counsellor virtually instead of always meeting in person?

[More details](#)



6. Which method of virtual communication would you prefer?

[More details](#)



7. How often do you use AI as a therapist?

[More details](#)

● Every time	2
● When I am under pressure	9
● I don't use AI at all	4



8. What do you think about the institution having a Counselling Services app?

13 Responses

ID ↑	Name	Responses
1	Andile Mokoena	It's a great idea since it will make mental health easy to reach for us students and stuff.
2	Matieho Mochekoane	For the time of the world we live in, that's a perfect idea .
3	Tshitso Selepe	I think it would help a lot of students who are unaware there is a counseling service at the university
4	Nthabiseng Melupe	That could also be a great idea
5	Innocentia Mokone	Good idea
6	Thembeke Mamadu	A counselling services app would help a lot of students because i think most of students fear judgement and fear having to tell a 'stranger' about their problems and actually having to open up having a counselling services app would make it seem as of i am talking to an AI as a therapist getting quick responses and not having my business heard by an actual person even though it may be the case
7	Jason Lesudi	Good option



9	Thabo Mohapi	It would be good thing to have
10	Katlego Shebe	that would be great
11	Bokang Semouse	I think it is a great idea because everyone can access it and assist those who are afraid of talking in front of others
12	Zizipho Zitha	That would be a good initiative considering the fact that it's always difficult for students to go and ask for counselling in person and some don't know where to email or whom to contact
13	Sherly Netshiavha	I think it's a brilliant idea, it will save the waiting time and make it easy for everyone to have access after hours.



PROJECT INFRASTRUCTURE

GITHUB REPOSITORY: <https://github.com/BoboNtsutle/Counselling-Services>

SUPABASE: <https://gslgnidvqcngwsnvhowv.supabase.co>

BoboNtsutle's Project

<https://gslgnidvqcngwsnvhowv.supabase.co>

Copy



STATUS
Healthy



COMPUTE
NANO



GITHUB
BoboNtsutle/Counselling-Servi...



RECENT BRANCH
No branches



LAST MIGRATION
No migrations



LAST BACKUP
No backups



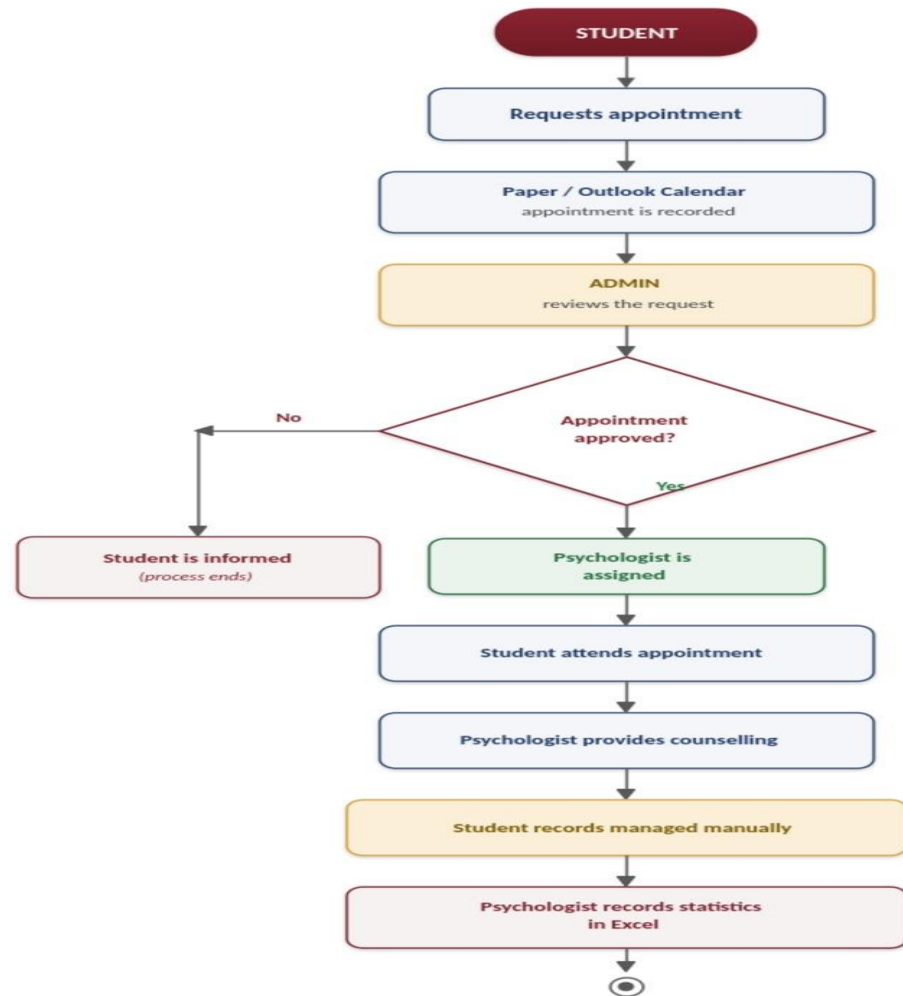
Primary Database
West EU (Ireland)
eu-west-1 · t3.nano



CPU 2% · Disk 14% · RAM 48% · 5/60 conns

Current Appointment Booking Process

Wellness Centre — as-is process (manual / Outlook-based)



Legend

Process step

Assignment step

Manual / paper-based step (pain point)

Terminal / outcome step

Decision point

SWOT ANALYSIS

Counselling Services Appointment Manager and Digital Communication App

DEV OPS — Group D

STRENGTHS	WEAKNESSES
• Directly targets well-documented, real problems (missed requests, double bookings, misplaced records) with clear stakeholder buy-in. • Cross-platform design (web + Flutter mobile) sharing one ASP.NET Core backend and Supabase database keeps data consistent across both apps. • Clear, well-defined scope with explicit out-of-scope items, reducing the risk of scope creep during development. • Built-in security focus: authentication, role-based access control, and encryption for student records are core requirements, not an afterthought. • Role separation (Students, Administrators, Psychologists/Counsellors) maps cleanly onto the real workflow.	• Six-person student team with no mention of prior large-scale team project experience increases coordination and integration risk. • Dual-platform delivery (web + mobile) with a shared backend roughly doubles front-end development and testing effort within a fixed timeline. • Reliance on third-party services (Supabase, and possibly a video-conferencing platform) introduces dependencies outside the team's control. • Handling confidential counselling and health-adjacent information raises the bar for data protection well above a typical student project. • Emergency/walk-in appointment handling and triage integration are complex edge cases easy to underestimate in early design.
OPPORTUNITIES	THREATS
• A successful system could be adopted operationally by the Wellness Centre beyond the academic deliverable, giving the project real-world impact. • The communication/announcement feature opens room to grow engagement (event promotion, awareness) beyond the core booking function. • Centralized statistics could later support data-driven decisions (e.g., staffing, peak demand periods) if reporting is extended. • Optional virtual counselling integration, if approved, could differentiate the system and extend its usefulness post-launch.	• Fixed academic deadlines could force cut corners on security or testing, especially risky for a system storing sensitive student data. • Any real or perceived mishandling of confidential counselling information could damage trust in the system, even after handover. • Change requests from the Wellness Centre (a real stakeholder) mid-project could push scope beyond what the team can deliver in time. • If double-booking prevention or triage logic has bugs, it reproduces the exact problem the system was built to solve.

Basis for this analysis

This SWOT is drawn from the group's Initial Software Requirements Specification: the problem statement, purpose, aim, objectives, scope, out-of-scope items, system constraints, and the planned web + Flutter/ASP.NET Core/Supabase architecture.

7. Functional Requirements (Updated)

Digital Communication removed

The following functional requirements are derived directly from the stakeholder needs. FR-08 (Digital Communication / virtual counselling) has been removed from the core functional requirements list and treated as a Could-Have item only, consistent with the refined MoSCoW prioritisation and stakeholder dependency on an approved video platform.

ID	Requirement	Description
FR-01	User Authentication	The system must allow authorised users to log in securely.
FR-02	Student Appointment Booking	The system must allow students to request counselling appointments online.
FR-03	Appointment Management	The system must allow admin staff to view, create, update, approve, reschedule and cancel appointments.
FR-04	Double-Booking Prevention	The system must prevent or flag double bookings of an appointment slot.
FR-05	Psychologist Appointment View	The system must allow psychologists to view their assigned appointments.
FR-06	Student Record Storage	The system must store student number, name, contact details and assigned psychologist.
FR-07	Record Search	The system must provide a search facility for authorised staff to locate student records.
FR-08	Appointment Status Updates	The system must allow the psychologist to update the status of an appointment.
FR-09	Role-Based Access Control	The system must restrict access to information and functions based on the user's role.
FR-10	Statistics Generation	The system must generate statistics and reports without requiring manual compilation.
FR-11	Announcements	The system must allow the Wellness Centre to publish programme and event announcements to students.

Note on removal

Former FR-08 (Digital Communication / virtual counselling) was removed from the Must-Have functional list. It remains a Could-Have item conditional on stakeholder approval of a video-conferencing platform.

10. Requirements Change Log — New Entry

The following change has been logged to record the removal of Digital Communication from core FRs:

Field	Value
Change ID	CR-013
Date Raised	2026-09-23
Requirement Affected	Functional Requirements — Digital Communication (former FR-08)
Change Type	Removal
Description of Change	Removed Digital Communication (virtual counselling with a counsellor) from the core Functional Requirements list. Virtual counselling remains a Could-Have item only.
Reason / Justification	Aligns the FR list with the refined MoSCoW prioritisation and the explicit dependency on stakeholder approval of a video-conferencing platform. Prevents treating an unapproved integration as essential.
Requested By	Group D (consistency review against Official SRS & refined scope)
Status	Implemented

Updated MoSCoW Alignment

With the removal of Digital Communication from the core FR list:

- Must Have remains focused on: registration/login, appointment booking & management, double-booking prevention, student records, RBAC, statistics (non-graphical), and basic staff–student communication (announcements / status messaging).
- Could Have continues to include: Virtual (video) counselling — conditional on stakeholder approval and an approved video-conferencing platform; Wellness tips; student feedback; customizable notifications.
- Won't Have is unchanged (medical diagnosis, prescribing, staff counselling via K-Ways, statistical graphs, etc.).
- The former FR-08 is therefore no longer listed among the numbered functional requirements that the system must implement as core deliverables.

Action required

Reflect this change in any subsequent design documents, the shared API contract, and the test plan.
Virtual counselling, if later approved, can be added back via a new change request.

1. Purpose

This section assesses whether the Counselling Services Appointment Manager and Digital Communication App is feasible to deliver as scoped, and identifies the main risks to delivering it successfully, along with mitigations. It is based on the group's Software Requirements Specification (problem statement, objectives, scope, constraints, assumptions, and planned architecture of Flutter mobile + ASP.NET Core web + shared Supabase backend).

2. Feasibility Analysis

Dimension	Assessment	Notes
Technical	Feasible	The chosen stack (Flutter, ASP.NET Core, Supabase/PostgreSQL) is well-documented, widely used, and matches skills typically covered in the IT curriculum. The main technical challenge is keeping the web and mobile apps in sync through one shared backend and API.
Operational	Feasible, with support needs	Wellness Centre staff and psychologists will need basic training and a simple, role-appropriate interface, since they are currently used to Outlook and paper-based workflows. Institutional buy-in (device/internet access, IT support) is assumed rather than guaranteed.
Economic	Feasible for an academic project	Development cost is primarily the team's time; cloud services (e.g., Supabase) typically have a usable free tier for a project of this scale. Any move to a production deployment for the actual Wellness Centre would need the institution to budget for hosting and maintenance.
Schedule	Feasible but tight	Delivering two client applications (web + mobile) plus a shared backend, security features, and reporting within an academic timeline is achievable only if scope is tightly prioritized and integration starts early rather than at the end.
Legal / Compliance	Feasible, requires care	Because the system stores confidential student and counselling-related information, it must be designed with South Africa's POPIA (Protection of Personal Information Act) principles in mind — lawful processing, access control, and safeguarding of personal information.

Overall feasibility

The project is feasible within an academic timeline provided the team prioritizes the core booking, record-management, and security requirements first, treats virtual counselling and advanced statistics as stretch goals, and begins integrating the web app, mobile app, and backend early rather than late in the project.

3. Risk Register

Each risk is scored using a standard 5-point scale for Likelihood and Impact (1 = Rare/Negligible, 3 = Possible/Moderate, 5 = Almost Certain/Severe). Risk Score = Likelihood × Impact (max 25). Scores of 15–25 are rated High priority, 8–14 Medium, and below 8 Low. Risks are listed from highest to lowest priority.

Score: 16 L=4 I=4 | Schedule

HIGH

Team members underestimate the effort of building and integrating both a web app and a Flutter mobile app on a shared backend.

Mitigation: Prioritize core appointment-booking and record features first; treat mobile and web as parallel workstreams with a shared API contract agreed early.

Score: 15 L=3 I=5 | Technical

HIGH

Double-booking prevention or triage logic contains bugs, reproducing the exact problem the system is meant to solve.

Mitigation: Write and prioritize automated tests for booking and triage logic; review this logic as a team before integration, not just before final submission.

Score: 15 L=3 I=5 | Security / Legal

HIGH

Confidential student/counselling data is exposed due to weak access control or unencrypted storage.

Mitigation: Apply role-based access control and encryption from the start (not retrofitted later); align data handling with POPIA principles for sensitive personal information.

Score: 12 L=3 I=4 | Quality

MEDIUM

Testing is left too late, so defects in booking, access control, or data handling are found close to the deadline.

Mitigation: Test each feature as it's built rather than only at the end; keep a running test/issue log from early in development.

Score: 9 L=3 I=3 | Team / Resource

MEDIUM

Uneven skill levels across the team in ASP.NET Core, Flutter, or Supabase slow down integration close to deadlines.

Mitigation: Assign work by current skill level, pair less-experienced members with stronger ones on shared modules, and start integration early rather than at the end.

Score: 9

L=3 I=3 | Scope

MEDIUM

Scope creep if the Wellness Centre (a real external stakeholder) requests additional features mid-project.

Mitigation: Keep the agreed scope and out-of-scope list visible to the client; log new requests as future work rather than folding them in immediately.

Score: 6

L=2 I=3 | Technical / External

LOW

Supabase or another third-party service has downtime, rate limits, or configuration issues the team can't control.

Mitigation: Build with the free/current tier's limits in mind, keep local test data separate from any shared instance, and confirm service status before demos.

Score: 6

L=3 I=2 | Scope / Schedule

LOW

Virtual counselling (video) integration is attempted but not completed in time, since it depends on stakeholder approval.

Mitigation: Treat virtual counselling as an optional stretch goal, clearly separated from the core deliverables that must ship.

Risk summary

Of the 8 risks identified: 3 rated High priority (score 15+), 3 rated Medium (score 8–14), and 2 rated Low (score below 8). High-priority risks should be reviewed by the team weekly; Medium risks at each milestone.

4. Key Recommendations

- 1 Lock down the shared API contract between the web app, mobile app, and ASP.NET Core backend early so both front ends can be built in parallel without blocking each other.
- 2 Build role-based access control and data protection in from the first version of the system, not as a late addition, given the sensitivity of counselling records.
- 3 Treat double-booking prevention and triage handling as priority features to test thoroughly, since they map directly to the problems the project is meant to solve.
- 4 Keep virtual counselling and advanced reporting/graphing as optional stretch goals so the core deliverables are not put at risk.
- 5 Start integration testing across web, mobile, and backend well before the final deadline rather than treating it as a last step.